

CUSTOMER CHURN ANALYSIS REPORT

Prepared By: Suditi Singh

Project: Customer Churn Analysis

Tools Used: Jupyter Notebook (Pandas), PostgreSQL, Power BI

Executive Summary

An end-to-end churn analysis was conducted on **15,000 customers** to identify churn drivers and revenue risk areas.

● Key Finding:

The overall **churn rate is 55%**, meaning more than half of the customers are leaving or at high risk.

This level of churn represents a significant business and revenue concern.

Methodology

Step 1: Data Cleaning – Jupyter Notebook (Pandas)

- Handled missing values (region, login frequency, operation medium, etc.)
- Replaced inconsistent values (e.g., “?” in referral column)
- Created new columns such as:
 - Tenure Days
 - Joining Month
 - Joining Year
- Exported cleaned dataset for database analysis

Step 2: Data Processing – PostgreSQL

- Imported cleaned dataset
- Performed aggregations:
 - Total customers
 - Total churned
 - Churn rate calculation
 - Complaints vs churn
 - Membership-wise churn
 - Region-based analysis

Step 3: Visualization – Power BI

- Created KPI dashboard
- Developed DAX measures for:
 - Total Customers

- Total Churned
 - Churn Rate %
 - High vs Low Value segmentation
- Built interactive visuals for business insights

Key Business Metrics

KPI	Value
Total Customers	15,000
Total Churned	8,000
Churn Rate	55%
Total Transaction Value ₹0.44 Billion	

Membership-Based Churn Analysis

Findings:

- Highest churn in **Basic Membership**
- High churn in **No Membership**
- Lower churn in **Gold and Silver Membership**

Business Insight:

Customers with lower engagement levels are more likely to churn.

Recommendation:

- Introduce loyalty rewards for Basic/No members
- Encourage upgrade to premium memberships
- Offer retention incentives

Regional Analysis

Findings:

- Majority of churn occurs in **City category**
- Rural/Town segments comparatively lower

Insight:

Urban customers form the largest risk segment.

Recommendation:

- Targeted marketing campaigns for city customers
- Behavioral segmentation within urban regions

Complaints vs Churn

Finding:

Churn distribution between customers with and without complaints is similar.

Insight:

Complaints are not the only driver of churn.
Engagement and value perception may be stronger factors.

High vs Low Value Customer Churn

Finding:

Significant churn observed among high-value customers.

Risk:

Direct revenue impact.

Recommendation:

- Early identification of high-value at-risk customers
- Personalized retention offers
- Dedicated support programs

Monthly Churn Trend

Peak Months:

- Month 7
- Month 12

Lowest Month:

- Month 2

Insight:

Possible seasonal churn pattern.

Recommendation:

- Pre-emptive retention campaigns before peak months
- Mid-year engagement strategies

Overall Business Risk Assessment

- 55% churn rate is critically high
- Revenue loss from high-value customers
- Weak retention in lower membership tiers

- Need for proactive engagement strategy

Strategic Action Plan

Immediate Actions:

- Protect high-value customers
- Improve engagement for Basic members
- Launch targeted retention campaigns

Long-Term Strategy:

- Develop predictive churn monitoring
- Implement personalized marketing
- Strengthen loyalty programs

Conclusion

This analysis highlights critical churn drivers and revenue risks.

By implementing targeted retention strategies and enhancing customer engagement, churn rate can be significantly reduced, improving long-term profitability and customer loyalty.